Lab 03: Authentication and Validation

Total: 10 points

Instructions

- Continue the Express project and Lab1 database from Lab 01 and Lab 02.
- Use environment variables for PORT, MONGODB_URI, and JWT_SECRET. Do not hardcode these values.
- Use the existing suppliers, products, and managers collections.
- Store manager passwords as bcrypt hashes in the pass field. Convert existing plaintext passwords to hashes.
- Apply server-side validation using express-validator.

Task 01 - Lab 01: JWT Authentication and API Validation - 5 points

1.1 JWT Authentication - 2 points

- Create POST /auth/jwt, accepting code and pass. Both fields are required.
- Find the manager by code and use bcrypt.compare() to verify the password.
- After successful login, return a JWT containing the manager's ID and an expiration time.
- Return a clear error for missing inputs or incorrect credentials.
- Protect all supplier and product APIs from Lab 01 using JWT authentication.
- Return 401 when the token is missing, invalid, or expired.

Marking criteria	Points
Login input validation and bcrypt password verification	0.5
Return a JWT containing the manager's ID and expiration	0.5
Protect all Lab 01 supplier and product APIs	1.0

Task 01 - continued

1.2 Validate Product Creation and Updates - 2 points

Validate the product create and update APIs from Lab 01.

All fields below are required.

Field	Validation requirement
name	Cannot contain only spaces.
category	Must be Laptop, Phone, or Accessory.
price	Must be a number greater than or equal to 0.
stock	Must be an integer greater than or equal to 0.
supplier	Must be a valid ID referencing an existing supplier.
tags	Must be an array of nonempty strings.
active	Must be a Boolean: true or false.

Return 400 with clear JSON validation errors. Invalid data must not be saved.

Marking criteria	Points
Validate all listed fields for product creation	0.75
Validate all listed fields for product updates	0.75
Return clear JSON errors and prevent invalid data from being saved	0.5

1.3 Save the Manager ID from the JWT - 1 point

- Add a manager field to the product model that references the managers collection.
- When creating or updating a product, obtain the manager's ID from the verified JWT and save it in this field. Do not hardcode it or accept it from the request body.
- On creation, this field identifies the creator. On update, it identifies the manager who last updated the product. Keep the supplier reference separate.

Marking criteria	Points
Save the manager ID from the verified JWT when creating a product	0.5
Save the manager ID from the verified JWT when updating a product	0.5

Task 02 - Lab 02: Login and Product Form Validation - 5 points

2.1 Validate Manager Login and Protect Pages - 2 points

- Both code and pass are required.
- Find the manager by code and use bcrypt.compare() to verify the password.
- Display an error for missing fields or incorrect credentials.
- After successful login, redirect to the product list.
- Require login for all product management pages and create, update, and delete actions.
- If the manager is not logged in, redirect to /auth/view.

Marking criteria	Points
Validate required code and pass, and display errors	0.5
Verify the password using bcrypt and redirect after successful login	0.5
Protect all management pages and actions; redirect unauthenticated requests	1.0

2.2 Validate All Product Fields for Create and Update - 3 points

All fields below are required.

Field	Validation requirement
name	Cannot contain only spaces.
category	Must be Laptop, Phone, or Accessory.
price	Must be a number greater than or equal to 0.
stock	Must be an integer greater than or equal to 0.
supplier	Must reference an existing supplier selected from the dropdown.
tags	Must be a string.
active	Convert the form value to a Boolean and save it correctly as true or false.

For example, tags entered as a,b,c must be saved as ["a", "b", "c"].

When validation fails:

- Do not save the product.
- Display errors near the relevant fields.
- Preserve entered values, including 0 for price or stock, the selected supplier, tags, and active status.
- Reopen the create modal or keep the edit form visible without crashing.

For create, update, and delete actions:

- Show a notification for both success and failure.
- After success, redirect to the product list.
- After failure, display the error on the current page or form.

Marking criteria	Points
Validate all listed fields for product creation	0.75
Validate all listed fields for product updates	0.75
Display field errors, preserve values, and keep the modal or form visible	1.0
Show success/error notifications and apply the required redirects	0.5